

Cox

24 Eros

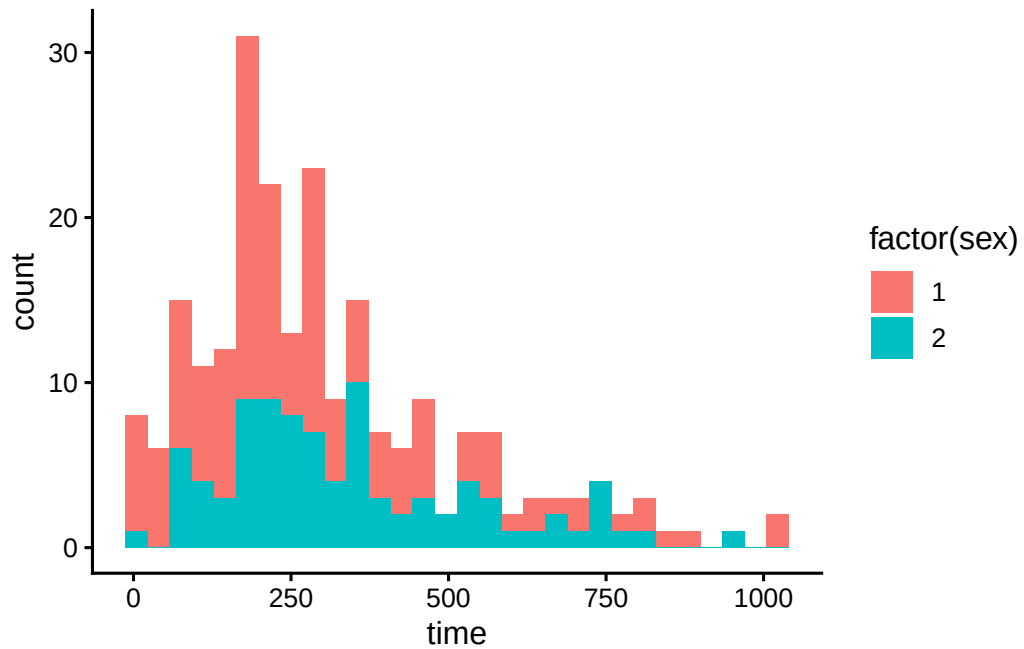
2026-06-09

```
library(tidyverse)
library(naniar)
library(patchwork)
library(survival)
library(ggpubr)
library(survminer)
library(mice)
library(gridExtra)
library(scales)
```

```
lung %>% head()
```

	inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
1	3	306	2	74	1	1	90	100	1175	NA
2	3	455	2	68	1	0	90	90	1225	15
3	3	1010	1	56	1	0	90	90	NA	15
4	5	210	2	57	1	1	90	60	1150	11
5	1	883	2	60	1	0	100	90	NA	0
6	12	1022	1	74	1	1	50	80	513	0

```
lung %>%
  ggplot(
    aes(
      x=time, fill=factor(sex)
    )
  )+
  geom_histogram()+
  theme_classic2()
```



•

data transform

0/1 1/2.

```
lung_clean <-
  lung %>%
  mutate(
    status=if_else(status==2,1,0)
    #status = recode(status, `1` = 0, `2` = 1)
  ) %>%
  as.tibble()
```

```

time
status
age
sex
inst          1-16
ph.ecog       ECOG    0-3
ph.karno      Karnofsky 0-100
pat.karno     Karnofsky
meal.cal
wt.loss       6

```

```
lung_clean %>%
  miss_var_summary()
```

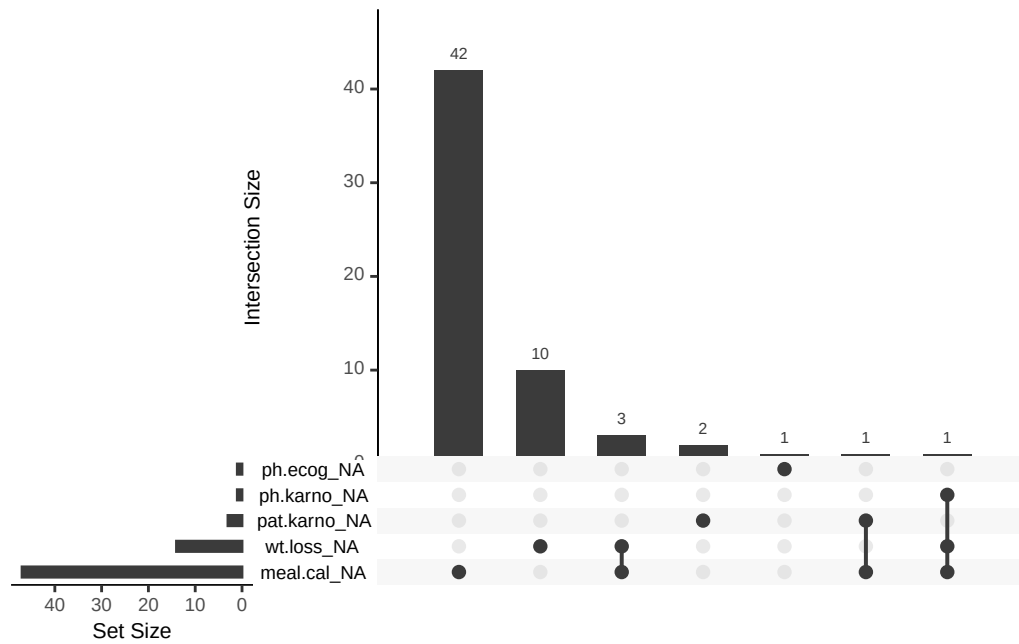
```

# A tibble: 10 x 3
  variable n_miss pct_miss
  <chr>    <int>    <num>
1 meal.cal      47    20.6
2 wt.loss      14     6.14
3 pat.karno      3     1.32
4 inst          1     0.439
5 ph.ecog        1     0.439
6 ph.karno        1     0.439
7 time          0      0
8 status         0      0
9 age           0      0
10 sex           0      0

```

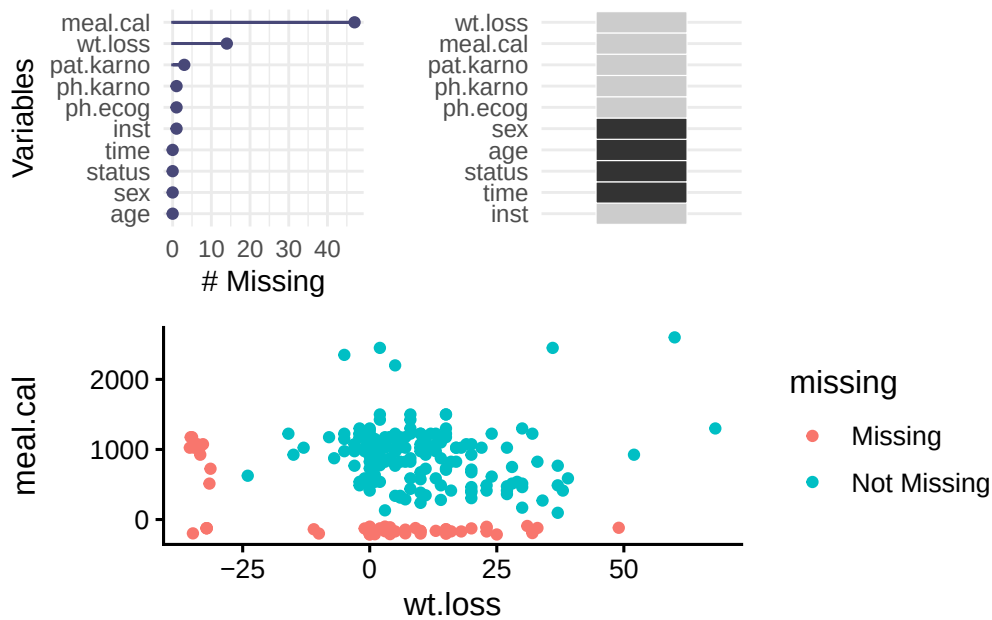
- meal.cal 20.6% MNAR
- wt.loss 6 14 MAR mice
-

```
lung_clean %>%
  gg_miss_upset()
```



```
p1 <-
  lung_clean %>%
  gg_miss_var()
p2 <-
  lung_clean %>%
  gg_miss_which()
p3 <-
  lung_clean %>%
  ggplot(aes(x=wt.loss, y=meal.cal)) +
  geom_miss_point() + theme_classic2()

(p1|p2)/p3
```



- meal.cal wt.loss 6
- wt.loss “ wt.loss / ”

Little's missing completely at random (MCAR) test

```
lung_clean %>%
  mcar_test()
```

```
# A tibble: 1 x 4
  statistic    df p.value missing.patterns
    <dbl> <dbl>   <dbl>         <int>
1     62.0    68  0.683             9
```

statistic	61.97617	Little	p
df	68		
p.value	0.6826042	p	
missing.patterns	9		

- **H** MCAR
- **H** MCAR MAR MNAR
- 0.68 meal.cal

```
lung_clean <-
lung_clean %>%
  mutate(
    is_na=recode(
      meal.cal,
      .missing=" ",
      .default=" ")
  )
```

```
fit <- survfit(
  formula = Surv(time, status) ~ is_na,
  data = lung_clean
)
```

```
sv_p1 <-
ggsurvplot(
  fit,
  pval = T, #p
  conf.int = T, #
  risk.table = TRUE,
  risk.table.col = "strata", #
  linetype = "strata", #
  surv.median.line = "hv", #
  ggtheme = theme_classic(), # ggplot2
  palette = c("#E7B800", "#2E9FDF"),
  legend="none",
  title=" "
)
```

```
sv_p2 <-
ggsurvplot(
  fit,
  #pval = T, #p
  conf.int = T, #
  #risk.table = TRUE,
  risk.table.col = "strata", #
  linetype = "strata", #
  surv.median.line = "hv", #
  ggtheme = theme_classic(), # ggplot2
```

```

palette = c("#E7B800", "#2E9FDF"),
legend = "none",
fun ="event",
title=" ")

sv_p3 <-
  ggsvrplot(
    fit,
    #pval = T, #p
    conf.int = T, #
    #risk.table = TRUE,
    risk.table.col = "strata", #
    linetype = "strata", #
    surv.median.line = "hv", #
    ggtheme = theme_classic(), # ggplot2
    palette = c("#E7B800", "#2E9FDF"),
    legend = "none",
    fun ="cumhaz",
    title=" ")

# -----
# 1      +
# -----
get_legend <- function(ggplot_obj) {
  tmp <- ggplot_gtable(ggplot_build(ggplot_obj))
  leg <- which(sapply(tmp$grobs, function(x) x$name) == "guide-box")
  tmp$grobs[[leg]]
}

#
legend_plot <- sv_p1$plot +
  labs(color = " ", linetype = " ") + #
  scale_color_manual(values = c("#E7B800", "#2E9FDF"), labels = c(" ", " ")) + #
  scale_linetype_manual(values = c("solid", "dashed"), labels = c(" ", " "))
legend <- get_legend(legend_plot + theme(legend.position = "bottom",
                                         legend.key.width = unit(1.5, "cm"))) #

# -----
# 2 sv_p1  +
# -----
sv_p1_complete <- sv_p1$plot / sv_p1$table +

```

```

plot_layout(heights = c(0.8, 0.2)) # : = 4:1

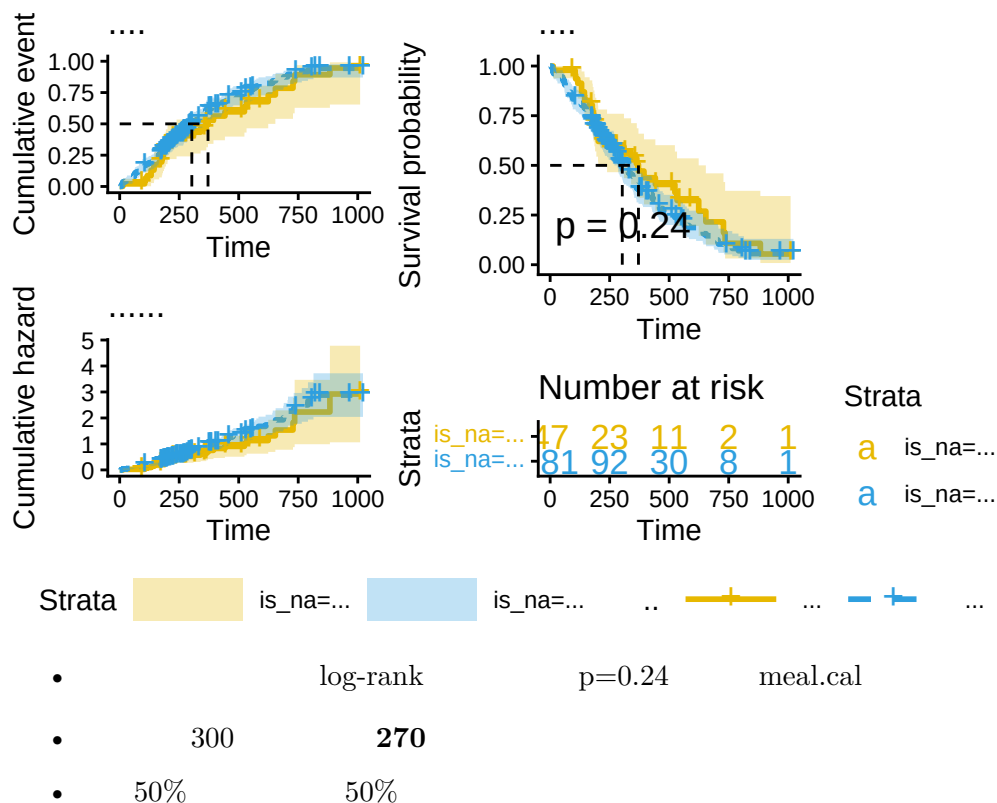
# -----
# 3 +
# -----
#
left_col <- sv_p2$plot / sv_p3$plot +
  plot_layout(heights = c(0.5, 0.5)) # 1:1

#
top_row <- left_col | sv_p1_complete +
  plot_layout(widths = c(0.5, 0.5)) # 1:1

#
final_plot <- (top_row / legend) +
  plot_layout(heights = c(0.9, 0.1)) # : = 9:1

final_plot

```



0. pmm
1. / U rf CART cart PMM
2. PMM → logreg → polyreg → polr
3. p » n PMM rf lda
4. → midastouch PMM → mice 2l.norm

- meal.cal, wt.loss pmm
- sex, ph.ecog logreg/

```
imp <- mice(
  lung_clean,
  method = "pmm", # PMM
  m = 5,
  maxit = 15, #
  printFlag = FALSE
)

m1 <-
bwplot(
  imp,
  ~ meal.cal + wt.loss,
  scales = list(x = list(relation = "free")),
  col = c("#377eb8", "#e41a1c"), #
  main = " ",
  ylab = " "
)

m2 <-
densityplot(
  imp,
  ~ meal.cal + wt.loss,
```

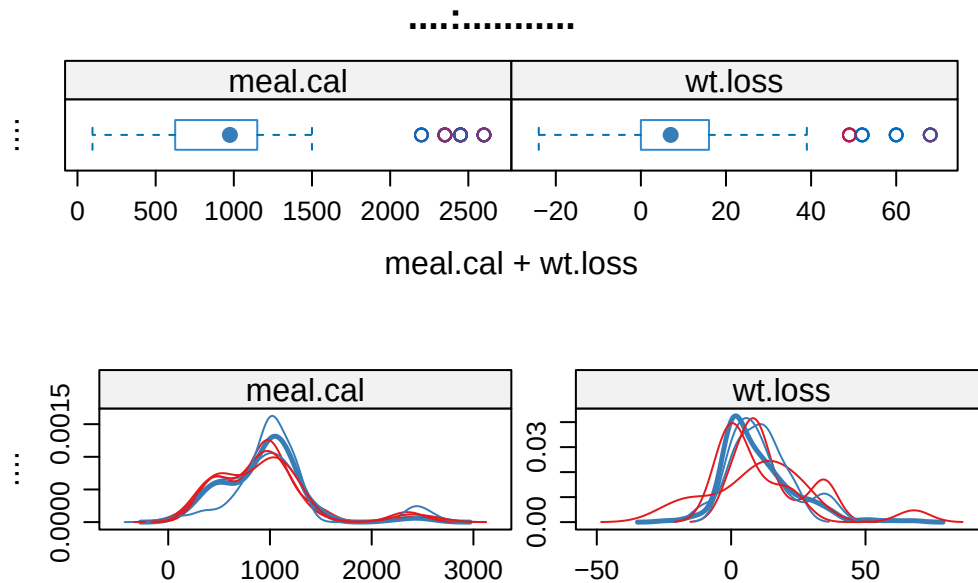
```

col = c("#377eb8", "#e41a1c"), #
main = NULL,
ylab = " "
)

m3 <- plot(
  imp,
  c("meal.cal", "wt.loss"),
  main = "MICE"
)

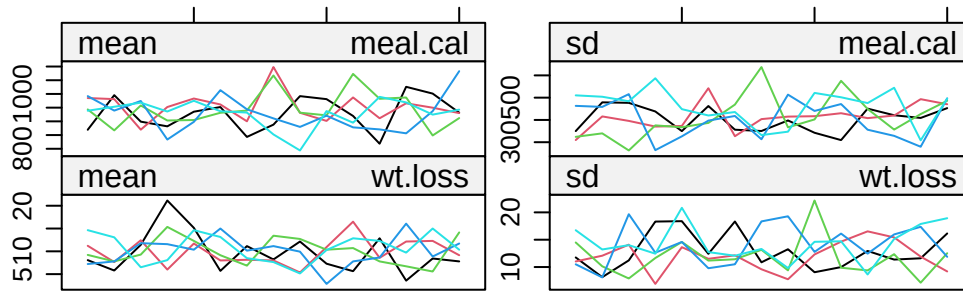
grid.arrange(
  m1, m2,
  nrow = 2,
  heights = c(2, 2)
)

```



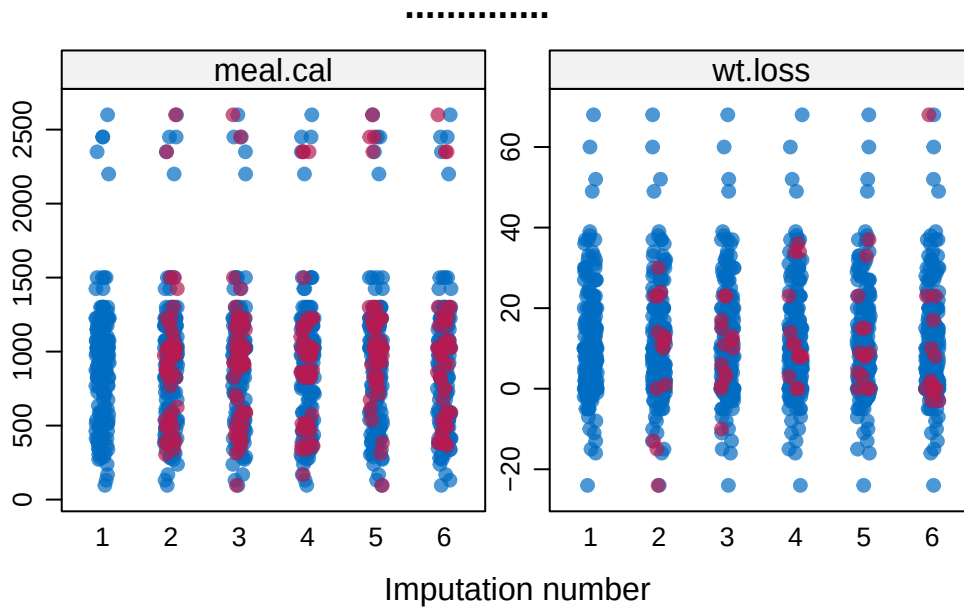
m3

MICE



Iteration

```
m4 <-
stripplot(
  imp,
  meal.cal + wt.loss ~ .imp,
  pch = 20,
  cex = 1.2,
  main = "          "
)
m4
```



- /
- meal.cal wt.loss
- meal.cal wt.loss
-

```
xy1 <-
  xyplot(
    imp,
    meal.cal ~ age,
    pch = 20,
    cex = 1.1,
    main = "meal.cal age "
  )

xy2 <-
  xyplot(
    imp,
    wt.loss ~ age,
    pch = 20,
    cex = 1.1,
    main = "wt.loss age "
  )

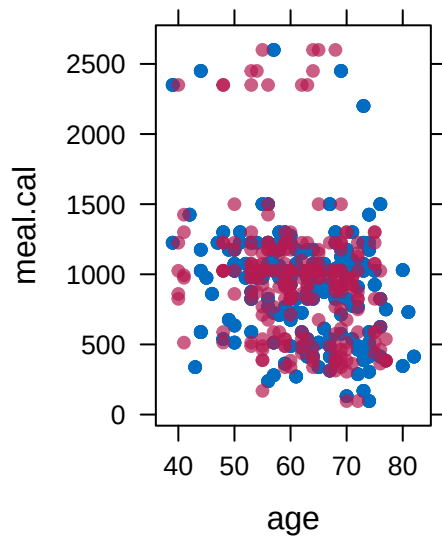
grid.arrange(
```

```

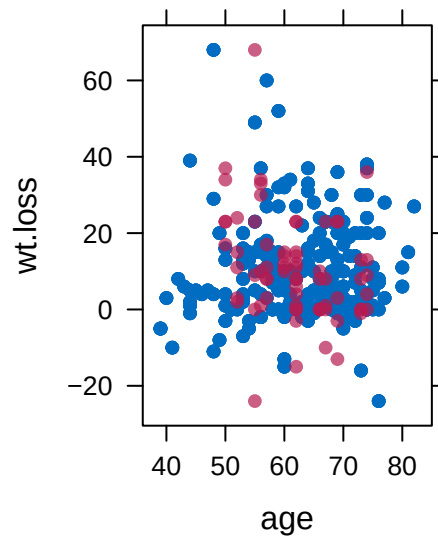
xy1, xy2,
ncol = 2
)

```

meal.cal . age



wt.loss . age



- wt.loss
- meal.cal

wt.loss
meal.cal age

MICE-PMM meal.cal wt.loss
wt.loss

meal.cal

COX

COX

```

fit_imp <- with(
  imp,
  coxph(Surv(time, status) ~ meal.cal + wt.loss + age + sex + ph.karno + ph.ecog)
)

pool_result <- pool(fit_imp)

```

```
summary(pool_result)
```

	term	estimate	std.error	statistic	df	p.value
1	meal.cal	-5.057225e-05	0.0002778551	-0.1820094	17.67109	0.8576469126
2	wt.loss	-9.042996e-03	0.0067027093	-1.3491553	111.93355	0.1800099217
3	age	1.122052e-02	0.0098435229	1.1398882	134.81241	0.2563540858
4	sex	-6.107382e-01	0.1748515989	-3.4928946	145.99485	0.0006326109
5	ph.karno	1.210083e-02	0.0093129473	1.2993556	156.54668	0.1957322086
6	ph.ecog	7.049665e-01	0.1811989810	3.8905653	155.38765	0.0001479708

COX

```
fit_cc <- coxph(
  Surv(time, status) ~ meal.cal + wt.loss + age + sex + ph.karno + ph.ecog,
  data = lung_clean,
  na.action = na.omit
)

summary(fit_cc)
```

Call:

```
coxph(formula = Surv(time, status) ~ meal.cal + wt.loss + age +
      sex + ph.karno + ph.ecog, data = lung_clean, na.action = na.omit)
```

```
n= 170, number of events= 123
(   58   )
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
meal.cal	-5.028e-05	9.999e-01	2.570e-04	-0.196	0.844890
wt.loss	-1.136e-02	9.887e-01	7.573e-03	-1.501	0.133439
age	1.346e-02	1.014e+00	1.157e-02	1.163	0.244805
sex	-5.570e-01	5.729e-01	1.986e-01	-2.805	0.005039 **
ph.karno	1.985e-02	1.020e+00	1.097e-02	1.809	0.070381 .
ph.ecog	7.943e-01	2.213e+00	2.129e-01	3.731	0.000191 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
meal.cal	0.9999	1.0001	0.9994	1.0005

wt.loss	0.9887	1.0114	0.9741	1.0035
age	1.0135	0.9866	0.9908	1.0368
sex	0.5729	1.7454	0.3882	0.8456
ph.karno	1.0201	0.9803	0.9983	1.0422
ph.ecog	2.2129	0.4519	1.4580	3.3588

Concordance= 0.636 (se = 0.029)

Likelihood ratio test= 25.78 on 6 df, p=2e-04

Wald test = 24.93 on 6 df, p=4e-04

Score (logrank) test = 25.43 on 6 df, p=3e-04

Cox ph.ecog	pooled Cox ECOG	sex	sex meal.cal	ph.ecog wt.loss	age
58			MICE-PMM		

```
# imp tibble
lung_imp1 <- complete(imp, 1) |>
  as_tibble()
```

cox PH

HR

-
-
- Cox

```
ph_results <- map_dfr(1:imp$m, function(i) {
  dat_i <- complete(imp, i)
  fit_i <- coxph(
    Surv(time, status) ~ meal.cal + wt.loss + age + sex + ph.karno + ph.ecog,
    data = dat_i
  )
  ph_i <- cox.zph(fit_i)
```

```

as.data.frame(ph_i$table) %>%
  rownames_to_column("term") %>%
  mutate(.imp = i) %>%
  select(.imp, term, chisq, df, p)
})

```

ph_results

	.imp	term	chisq	df	p
1	1	meal.cal	4.95672276	1	0.025989418
2	1	wt.loss	0.09591023	1	0.756793846
3	1	age	0.06084736	1	0.805161626
4	1	sex	2.52704067	1	0.111909921
5	1	ph.karno	6.20050285	1	0.012771402
6	1	ph.ecog	1.73205880	1	0.188148183
7	1	GLOBAL	15.03004257	6	0.020024334
8	2	meal.cal	4.58789766	1	0.032198485
9	2	wt.loss	0.06741135	1	0.795143822
10	2	age	0.07169398	1	0.788886133
11	2	sex	2.29216298	1	0.130028605
12	2	ph.karno	6.10098190	1	0.013510679
13	2	ph.ecog	1.77749380	1	0.182457375
14	2	GLOBAL	14.37068331	6	0.025758675
15	3	meal.cal	7.31332640	1	0.006844511
16	3	wt.loss	0.05072414	1	0.821807967
17	3	age	0.04603579	1	0.830110747
18	3	sex	2.23584312	1	0.134843054
19	3	ph.karno	5.60194488	1	0.017940551
20	3	ph.ecog	1.56382602	1	0.211105896
21	3	GLOBAL	15.69548975	6	0.015485154
22	4	meal.cal	10.19509840	1	0.001408144
23	4	wt.loss	0.11690449	1	0.732416381
24	4	age	0.06728095	1	0.795337653
25	4	sex	2.57301725	1	0.108700225
26	4	ph.karno	6.01995526	1	0.014145005
27	4	ph.ecog	1.76438083	1	0.184079027
28	4	GLOBAL	20.27135456	6	0.002477552
29	5	meal.cal	5.87566465	1	0.015351576
30	5	wt.loss	0.55691139	1	0.455507376
31	5	age	0.04720119	1	0.828007010
32	5	sex	2.00682257	1	0.156592990

```

33    5 ph.karno  5.78062161  1 0.016203810
34    5 ph.ecog  1.66357742  1 0.197121000
35    5 GLOBAL 15.18753459  6 0.018847208

```

```

ph_summary <- ph_results %>%
  group_by(term) %>%
  summarise(
    min_p = min(p, na.rm = TRUE),
    median_p = median(p, na.rm = TRUE),
    max_p = max(p, na.rm = TRUE),
    n_sig = sum(p < 0.05, na.rm = TRUE),
    .groups = "drop"
  )

ph_summary

```

```

# A tibble: 7 x 5
  term      min_p median_p max_p n_sig
<chr>    <dbl>    <dbl>  <dbl> <int>
1 GLOBAL  0.00248    0.0188 0.0258     5
2 age      0.789    0.805  0.830     0
3 meal.cal 0.00141    0.0154 0.0322     5
4 ph.ecog  0.182    0.188  0.211     0
5 ph.karno 0.0128    0.0141 0.0179     5
6 sex      0.109    0.130  0.157     0
7 wt.loss  0.456    0.757  0.822     0

```

```

#      Schoenfeld      Cox      PH assumption
#      HR      PH
#      PH

```

- Cox GLOBAL 5 meal.cal
 ph.karno PH age wt.loss sex ph.ecog sex ph.ecog PH
 ph.karno ph.ecog ph.karno PH ph.karno ph.ecog

ph.karno

```

fit_imp_no_karno <- with(
  imp,
  coxph(Surv(time, status) ~ meal.cal + wt.loss + age + sex + ph.ecog)
)

pool_no_karno <- pool(fit_imp_no_karno)

summary(pool_no_karno)

```

	term	estimate	std.error	statistic	df	p.value
1	meal.cal	-5.543882e-05	0.0002739356	-0.2023790	18.09033	8.418790e-01
2	wt.loss	-8.745357e-03	0.0066119237	-1.3226645	122.08605	1.884193e-01
3	age	9.626687e-03	0.0097718559	0.9851441	133.17518	3.263397e-01
4	sex	-5.857847e-01	0.1737813457	-3.3708148	147.39046	9.570487e-04
5	ph.ecog	5.293791e-01	0.1209687254	4.3761648	156.44296	2.198378e-05

```

ph_results_no_karno <- map_dfr(1:imp$m, function(i) {

  dat_i <- complete(imp, i)

  fit_i <- coxph(
    Surv(time, status) ~ meal.cal + wt.loss + age + sex + ph.ecog,
    data = dat_i
  )

  ph_i <- cox.zph(fit_i)

  as.data.frame(ph_i$table) %>%
    rownames_to_column("term") %>%
    mutate(.imp = i) %>%
    select(.imp, term, chisq, df, p)
})

ph_summary_no_karno <- ph_results_no_karno %>%
  group_by(term) %>%
  summarise(
    min_p = min(p, na.rm = TRUE),
    median_p = median(p, na.rm = TRUE),
    max_p = max(p, na.rm = TRUE),
    n_sig = sum(p < 0.05, na.rm = TRUE),
    .groups = "drop"
  )

```

```
)
```

```
ph_summary_no_karno
```

```
# A tibble: 6 x 5
```

	term <chr>	min_p <dbl>	median_p <dbl>	max_p <dbl>	n_sig <int>
1	GLOBAL	0.00512	0.0488	0.0624	3
2	age	0.646	0.663	0.697	0
3	meal.cal	0.00133	0.0126	0.0287	5
4	ph.ecog	0.155	0.161	0.178	0
5	sex	0.0749	0.0911	0.113	0
6	wt.loss	0.505	0.823	0.881	0

- ph.karno PH meal.cal 5 PH
- ph.karno PH meal.cal

meal.cal

```
fit_imp_final2 <- with(
  imp,
  coxph(Surv(time, status) ~ wt.loss + age + sex + ph.ecog)
)

summary(pool(fit_imp_final2))
```

	term	estimate	std.error	statistic	df	p.value
1	wt.loss	-0.008323477	0.006467297	-1.287010	133.9970	2.003095e-01
2	age	0.010581908	0.009145848	1.157018	158.9826	2.490006e-01
3	sex	-0.578275568	0.168143422	-3.439180	158.7796	7.454011e-04
4	ph.ecog	0.527033358	0.120445755	4.375691	157.0138	2.198195e-05

```
ph_results_final2 <- map_dfr(1:imp$m, function(i) {

  dat_i <- complete(imp, i)

  fit_i <- coxph(
    Surv(time, status) ~ wt.loss + age + sex + ph.ecog,
```

```

    data = dat_i
  )

  ph_i <- cox.zph(fit_i)

  as.data.frame(ph_i$table) %>%
    rownames_to_column("term") %>%
    mutate(.imp = i) %>%
    select(.imp, term, chisq, df, p)
})

ph_summary_final2 <- ph_results_final2 %>%
  group_by(term) %>%
  summarise(
    min_p = min(p, na.rm = TRUE),
    median_p = median(p, na.rm = TRUE),
    max_p = max(p, na.rm = TRUE),
    n_sig = sum(p < 0.05, na.rm = TRUE),
    .groups = "drop"
  )

ph_summary_final2

```

A tibble: 5 x 5

	term	min_p	median_p	max_p	n_sig
	<chr>	<dbl>	<dbl>	<dbl>	<int>
1	GLOBAL	0.271	0.284	0.315	0
2	age	0.683	0.687	0.696	0
3	ph.ecog	0.158	0.168	0.174	0
4	sex	0.0897	0.0928	0.103	0
5	wt.loss	0.503	0.829	0.889	0

- | | | | | | | | | | | | | |
|-----|--------|-----|---------|-----|-----|---------|--|-----|----------|--|----------|-----|
| | Cox | | wt.loss | age | sex | ph.ecog | | sex | ph.ecog | | wt.loss | age |
| sex | lung | 1 = | 2 = | | | ph.ecog | | | ECOG | | | |
| | GLOBAL | | PH | | | Cox | | | meal.cal | | ph.karno | |

HR

```
summary(pool(fit_imp_final2), conf.int = TRUE, exponentiate = TRUE)
```

	term	estimate	std.error	statistic	df	p.value	2.5 %
1	wt.loss	0.9917111	0.006467297	-1.287010	133.9970	2.003095e-01	0.9791067
2	age	1.0106381	0.009145848	1.157018	158.9826	2.490006e-01	0.9925468
3	sex	0.5608647	0.168143422	-3.439180	158.7796	7.454011e-04	0.4023786
4	ph.ecog	1.6938997	0.120445755	4.375691	157.0138	2.198195e-05	1.3352658

	97.5 %	conf.low	conf.high
1	1.0044777	0.9791067	1.0044777
2	1.0290592	0.9925468	1.0290592
3	0.7817741	0.4023786	0.7817741
4	2.1488576	1.3352658	2.1488576

- 56.1%
- ECOG 1 69.4%

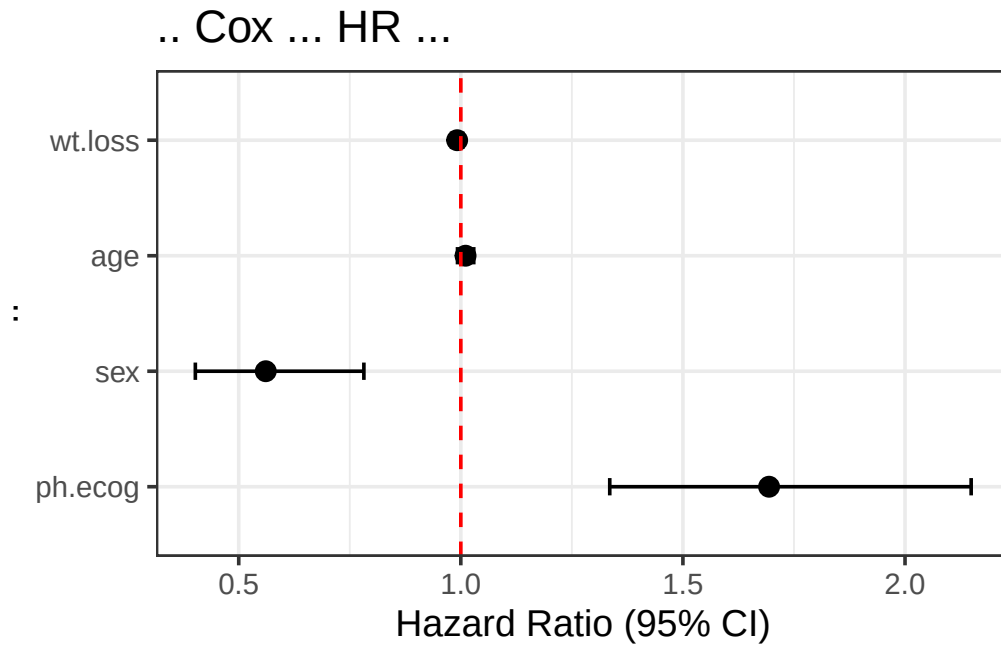
Cox forest plot

```
hr_table <- summary(pool(fit_imp_final2), conf.int = TRUE, exponentiate = TRUE) %>%
  transmute(
    term,
    HR = estimate,
    CI_low = `2.5 %`,
    CI_high = `97.5 %`,
    p.value
  ) %>%
  mutate(
    term = factor(term, levels = rev(c("wt.loss", "age", "sex", "ph.ecog")))
  )

ggplot(hr_table, aes(x = term, y = HR)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = CI_low, ymax = CI_high), width = 0.15) +
  geom_hline(yintercept = 1, linetype = 2, color = "red") +
  coord_flip() +
  labs(
    title = " Cox HR ",

```

```
x = " ",
y = "Hazard Ratio (95% CI)"
) +
theme_bw(base_size = 14)
```



- 1 HR < 1
- 1 HR > 1
- 1

Kaplan-Meier sex

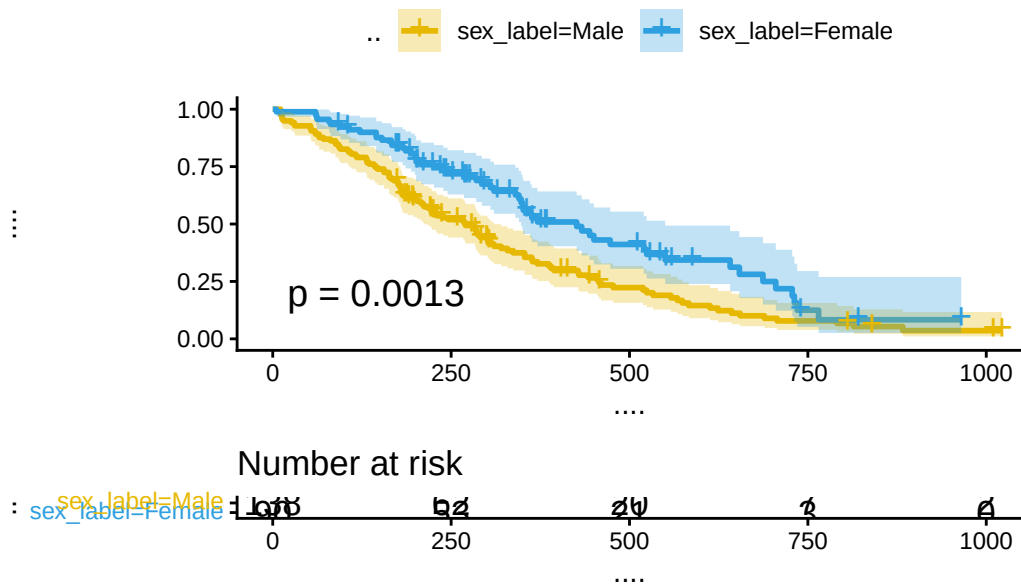
```
lung_imp1 <- complete(imp, 1) %>%
  mutate(
    sex_label = factor(sex, levels = c(1, 2), labels = c("Male", "Female"))
  )

fit_sex <- survfit(Surv(time, status) ~ sex_label, data = lung_imp1)

ggsurvplot(
```

```
fit_sex,
data = lung_imp1,
risk.table = TRUE,
pval = TRUE,
conf.int = TRUE,
palette = c("#E7B800", "#2E9FDF"),
ggtheme = theme_classic(),
title = "Kaplan-Meier",
xlab = "",
ylab = "",
legend.title = "")
```

..... Kaplan-Meier



Kaplan-Meier **ph.ecog**

```
lung_imp1 <- complete(imp, 1) %>%
  mutate(
    ecog_group = case_when(
      ph.ecog == 0 ~ "0",
      ph.ecog == 1 ~ "1",
      ph.ecog >= 2 ~ "2+",
      TRUE ~ NA_character_
```

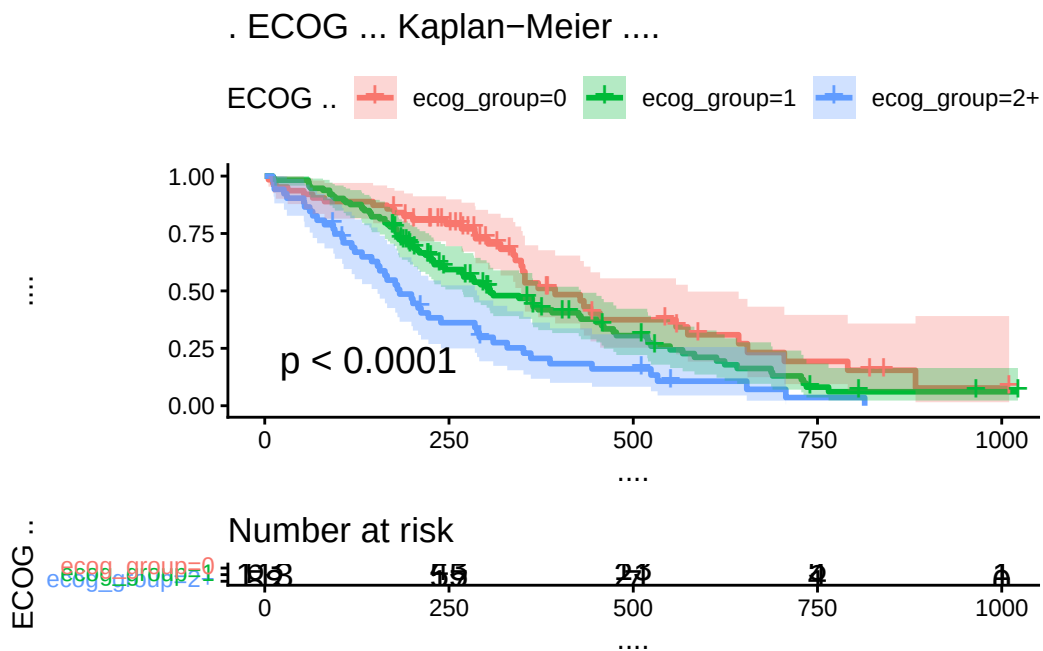
```

    ),
    ecog_group = factor(ecog_group, levels = c("0", "1", "2+"))
  )

fit_ecog <- survfit(Surv(time, status) ~ ecog_group, data = lung_imp1)

ggsurvplot(
  fit_ecog,
  data = lung_imp1,
  risk.table = TRUE,
  pval = TRUE,
  conf.int = TRUE,
  ggtheme = theme_classic(),
  title = " ECOG Kaplan-Meier ",
  xlab = " ",
  ylab = " ",
  legend.title = "ECOG "
)

```



PH

```
ph_plot_data <- ph_summary_final2 %>%
  mutate(
    term = factor(term, levels = rev(term)),
    label = paste0("n_sig = ", n_sig)
  )

ggplot(ph_plot_data, aes(y = term)) +
  geom_vline(
    xintercept = 0.05,
    linetype = 2,
    color = "red"
  ) +
  geom_linerange(
    aes(xmin = min_p, xmax = max_p),
    linewidth = 1
  ) +
  geom_point(
    aes(x = median_p),
    size = 3
  ) +
  geom_text(
    aes(x = max_p, label = label),
    hjust = -0.1,
    size = 4
  ) +
  labs(
    title = "      PH      ",
    subtitle = "      p      p  ",
    x = "PH  p  ",
    y = " "
  ) +
  coord_cartesian(xlim = c(0, max(ph_plot_data$max_p) * 1.2)) +
  theme_bw(base_size = 14)
```

